HDOC Day-44

String Manipulation Using a Menu-Driven C Program

Aim

To write a menu-driven C program using separate user-defined functions to reverse each word in a sentence without changing word order, print the initials of a name, and exit the program.

Algorithm

1. Start the program.

2. Display a menu: Reverse Each Word, Print Initials, and Exit.

3. Accept the user choice and use a switch statement to call the selected function.

4. For reversing each word, read a sentence and scan it character by character. Identify each word boundary, reverse only that word in place, and preserve the order of words and original spaces.

5. For initials, read a full name and scan each character. At the beginning of each name component, print the first alphabetic character in uppercase followed by a period.

6. Display an error message for an invalid choice.

7. Repeat the menu until Exit is selected, then stop the program.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 200

void reverseEachWord(void);
void printInitials(void);

int main()
{
    int choice;

    do
    {
        printf("\n----- STRING OPERATIONS MENU -----\n");
        printf("1. Reverse Each Word in a Sentence\n");
        printf("2. Print Initials of a Name\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1)
        {
            int ch;
            printf("Invalid input! Enter a number.\n");
            while ((ch = getchar()) != '\n' && ch != EOF) { }
            choice = 0;
            continue;
        }
        {
            int ch;
```

```
            while ((ch = getchar()) != '\n' && ch != EOF) { }
        }

        switch (choice)
        {
            case 1: reverseEachWord(); break;
            case 2: printInitials(); break;
            case 3: printf("Exiting program...\n"); break;
            default:
                printf("Invalid choice! Please enter 1 to 3.\n");
        }
    } while (choice != 3);

    return 0;
}

/* Reverse characters inside each word */
void reverseEachWord(void)
{
    char sentence[MAX], temp;
    int start = 0, i = 0, left, right;

    printf("Enter a sentence: ");
    if (fgets(sentence, MAX, stdin) == NULL) return;
    sentence[strcspn(sentence, "\n")] = '\0';

    while (1)
    {
        if (sentence[i] == ' ' || sentence[i] == '\0')
        {
            left = start;
            right = i - 1;
            while (left < right)
            {
                temp = sentence[left];
                sentence[left] = sentence[right];
                sentence[right] = temp;
                left++;
                right--;
            }
            if (sentence[i] == '\0') break;
            start = i + 1;
        }
        i++;
    }
    printf("Sentence after reversing each word: %s\n",
           sentence);
}

/* Print uppercase initials followed by periods */
void printInitials(void)
{
    char name[MAX];
    int i, newWord = 1;

    printf("Enter a full name: ");
    if (fgets(name, MAX, stdin) == NULL) return;
    name[strcspn(name, "\n")] = '\0';
    printf("Initials: ");

    for (i = 0; name[i] != '\0'; i++)
    {
        if (isspace((unsigned char)name[i]))
            newWord = 1;
        else if (newWord && isalpha((unsigned char)name[i]))
        {
```

```
            printf("%c.", toupper((unsigned char)name[i]));
            newWord = 0;
        }
    }
    printf("\n");
}
```

Test Case Table

Test	Operation	Input	Expected Result
1	Reverse Each Word	Hello World	olleH dlroW
2	Reverse Each Word	What is special?	tahW si ?laiceps
3	Reverse Each Word	C Programming	C gnimmargorP
4	Reverse Each Word	I love coding	I evol gnidoc
5	Print Initials	Alexander Graham Bell	A.G.B.
6	Print Initials	John Smith	J.S.
7	Print Initials	Rahul Kumar Sharma	R.K.S.
8	Print Initials	Alice	A.
9	Invalid Choice	Choice = 4	Invalid choice message
10	Exit	Choice = 3	Program exits

Sample Output

```
----- STRING OPERATIONS MENU -----
1. Reverse Each Word in a Sentence
2. Print Initials of a Name
3. Exit
Enter your choice: 1
Enter a sentence: What is special?
Sentence after reversing each word: tahW si ?laiceps

----- STRING OPERATIONS MENU -----
1. Reverse Each Word in a Sentence
2. Print Initials of a Name
3. Exit
Enter your choice: 2
Enter a full name: Alexander Graham Bell
Initials: A.G.B.

----- STRING OPERATIONS MENU -----
1. Reverse Each Word in a Sentence
2. Print Initials of a Name
3. Exit
Enter your choice: 4
Invalid choice! Please enter 1 to 3.

----- STRING OPERATIONS MENU -----
1. Reverse Each Word in a Sentence
2. Print Initials of a Name
3. Exit
Enter your choice: 3
```

```
Exiting program...
```

Result

The menu-driven C program was designed using separate user-defined functions, character arrays, loops, conditional statements, and a switch-case statement. It reverses each word independently while preserving word order and spacing, prints uppercase initials followed by periods, handles invalid menu choices, and exits when requested.